

## Assignment 3: Raw Socket Packet Analysis and Protocol Investigation

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### 1. Objective

The objective of this assignment is to understand Linux raw socket programming and IP packet structures by capturing TCP packets using raw sockets. The captured packets were analyzed by extracting IP header fields and TCP-specific fields and verified using Wireshark.

### 2. System Configuration

Operating System: Ubuntu Linux

Linux vivobookasuslaptop515jab 6.17.0-35-generic

#35~24.04.1-Ubuntu SMP PREEMPT\_DYNAMIC Tue May 26 19:30:42 UTC 2 x86\_64 x86\_64  
GNU/Linux

### 3. Compile and Run Commands

```
gcc raw_capture.c -o raw_capture
```

```
sudo ./raw_capture
```

```
sudo ./raw_capture > program_output.txt
```

### 4. Program Design

The program creates a TCP raw socket, captures packets using `recvfrom()`, extracts IP and TCP headers, displays Source IP, Destination IP, Protocol Number, TTL, Packet Size, Source Port, Destination Port, TCP Flags and IP Version, then stops after capturing 20 packets.

### 5. Traffic Generation

Protocol: TCP

Commands Used:

1. `sudo ./raw_capture`

2. `nc -l 5000`

3. `nc 127.0.0.1 5000`

Messages sent: hello, test, network, assignment, raw socket.

Interface: lo (Loopback).

```
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc -l
hello
test
raw socket
testing
raw socket
testing 123
hello
^C
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc -l
hello
test
network
raw socket
assignment
^C
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc -l
hello
test
network
raw socket
assignment
^C
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc -l
hello
test
network
raw socket
assignment
loopback l0 network
█
```

```
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc 127.0.0.1
hello
test
raw socket
testing
raw socket
testing 123
hello
^C
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc 127.0.0.1
hello
test
network
raw socket
assignment
^C
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc 127.0.0.1
hello
test
network
raw socket
assignment
^C
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc 127.0.0.1
hello
test
network
raw socket
assignment
^C
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ nc 127.0.0.1
hello
test
network
raw socket
assignment
loopback l0 network
█
```

```
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ ls
raw_capture raw_capture.c
sadhak@vivobookasuslaptopx515jab:~/Desktop/Assignment03$ sudo ./raw_capture
[sudo] password for sadhak:
ROLL_NO=241801168007
ASSIGNED_PROTOCOL=TCP
=====
PACKET_NO=1
SRC_IP=32.186.73.121
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=243
PACKET_SIZE=60
IP_VERSION=4
SRC_PORT=443
DST_PORT=51676
TCP_FLAGS=SYN ACK
=====
PACKET_NO=2
SRC_IP=52.32.228.63
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=243
PACKET_SIZE=60
IP_VERSION=4
SRC_PORT=443
DST_PORT=34968
TCP_FLAGS=SYN ACK
=====
PACKET_NO=3
SRC_IP=192.178.173.95
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=119
PACKET_SIZE=60
IP_VERSION=4
SRC_PORT=443
DST_PORT=48670
TCP_FLAGS=SYN ACK
=====
PACKET_NO=4
SRC_IP=192.178.173.95
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=120
PACKET_SIZE=60
IP_VERSION=4
SRC_PORT=443
DST_PORT=48668
TCP_FLAGS=SYN ACK
=====
PACKET_NO=5
SRC_IP=151.101.211.52
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=57
PACKET_SIZE=60
IP_VERSION=4
SRC_PORT=88
DST_PORT=33428
TCP_FLAGS=SYN ACK
=====
PACKET_NO=6
SRC_IP=151.101.211.52
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=57
PACKET_SIZE=52
IP_VERSION=4
SRC_PORT=88
DST_PORT=33428
TCP_FLAGS=ACK
=====
PACKET_NO=7
SRC_IP=151.101.211.52
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=57
PACKET_SIZE=348
IP_VERSION=4
SRC_PORT=88
DST_PORT=33428
TCP_FLAGS=ACK PSH
=====
PACKET_NO=8
SRC_IP=151.101.211.52
DST_IP=18.146.228.228
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=57
PACKET_SIZE=54
IP_VERSION=4
SRC_PORT=88
DST_PORT=33428
```

## 6. Experimental Results

The raw socket program successfully captured 20 TCP packets. Both localhost (127.0.0.1) traffic and external TCP traffic were observed. Refer to program\_output.txt for detailed packet information.

## 7. Wireshark Verification

| Frame | Source IP | Destination IP | Protocol | TTL | Packet Size | Matched |
|-------|-----------|----------------|----------|-----|-------------|---------|
| 2     | 127.0.0.1 | 127.0.0.1      | TCP      | 64  | 66 bytes    | Yes     |
| 3     | 127.0.0.1 | 127.0.0.1      | TCP      | 64  | 66 bytes    | Yes     |
| 4     | 127.0.0.1 | 127.0.0.1      | TCP      | 64  | 66 bytes    | Yes     |
| 5     | 127.0.0.1 | 127.0.0.1      | TCP      | 64  | 74 bytes    | Yes     |
| 6     | 127.0.0.1 | 127.0.0.1      | TCP      | 64  | 74 bytes    | Yes     |

## 8. Header Analysis

### 1. Why is the TTL value not zero?

The selected packets were captured on the loopback interface (127.0.0.1). Since they did not pass through any routers, the TTL value remained 64 instead of decreasing to zero.

### 2. Why do packet sizes differ?

Packet sizes differ because TCP control packets (ACK, SYN, FIN) and data packets carry different amounts of information. SYN packets include additional TCP options, while data packets include application payloads.

### 3. What information does the Protocol field provide?

The Protocol field identifies the transport-layer protocol encapsulated inside the IP packet. In this experiment, the value was 6, indicating TCP.

### 4. What could happen if the Protocol field is modified?

The receiving system may interpret the packet incorrectly or discard it because the protocol number no longer matches the packet format.

### 5. What does the protocol-specific field reveal?

The TCP source port, destination port, and TCP flags identify the communicating applications and the state of the TCP connection, such as connection establishment, data transfer, acknowledgement, or termination.

## 9. Program Enhancement

Additional IP header field displayed: IP Version.

Observed value: 4 for all sampled packets. This indicates all captured packets used IPv4.

## 10. Reflection Answers

### 1. Why are root privileges required for raw sockets?

Raw sockets provide direct access to packet headers and network traffic. Root privileges are required to prevent unauthorized users from capturing or modifying packets and to protect system security.

### 2. How are raw sockets different from TCP/UDP sockets?

Raw sockets provide access to complete packets including IP and TCP headers, whereas TCP/UDP sockets provide only application data after the operating system processes the transport layer.

### 3. State one advantage and one limitation of raw sockets.

Advantage: They allow detailed packet inspection and protocol analysis.

Limitation: They require administrator privileges and are more difficult to program than standard sockets.

### 4. Mention one networking or cybersecurity application of raw sockets.

Raw sockets are used in packet analyzers, intrusion detection systems (IDS), and network troubleshooting tools.

## 11. Conclusion

The assignment successfully demonstrated TCP packet capture using Linux raw sockets. Packet fields extracted by the program matched the values observed in Wireshark, improving understanding of IP and TCP headers and packet analysis.